ParaGen: Controllable Neural Paraphrase Generation with Semantic Preservation

Aman Mujahid **Harsh Toshniwal** **Shreya Shastry** **Tianyu Zhang** **Vanshika**

{amujahid, htoshniv, skshastr, tzhang44, v567458}@usc.edu

Abstract

Paraphrase generation (rewriting text while preserving semantic meaning) is fundamental to NLP applications including data augmentation, query rewriting, and text simplification. While transformer models show promise, they struggle with semantic fidelity and output diversity. We propose ParaGen, a neural paraphrase generation system that extends fine-tuned T5 with controllable generation techniques focused on length and lexical diversity control. The system generates high-quality, semantically-preserving paraphrases, augmented with a semantic reranking step that selects candidates based on embedding similarity. Our work reproduces and extends the controllable paraphrase generation paradigm from recent literature [1]. We evaluate our approach using both automatic metrics and human evaluation on standard benchmark datasets.

1 Introduction

Paraphrase generation is the task of automatically producing alternative expressions of a given text that convey the same semantic meaning while varying in surface form. This capability is essential across diverse NLP applications, from enhancing search systems through query reformulation to augmenting training data for downstream tasks and improving text accessibility through simplification.

Despite significant advances in neural language models, paraphrase generation remains challenging. The primary objective is to balance two competing constraints: semantic preservation, which ensures the generated paraphrase maintains the original meaning, and lexical or syntactic diversity, which produces outputs that are sufficiently different from the input to be useful. Existing approaches often bias toward either conservative outputs that closely mimic the source, achieving high semantic similarity but low diversity, or creative variations that risk semantic drift.

Moreover, many real-world applications require controllable paraphrase generation, where users can specify desired attributes such as sentence length or degree of lexical variation. Current end-to-end models typically lack explicit mechanisms for such control, limiting their practical utility in diverse application contexts.

1.1 Problem Definition

The task takes as input a source sentence $s = \{w_1, w_2, ..., w_n\}$ in natural language, and produces as output a target paraphrase $t = \{w'_1, w'_2, ..., w'_m\}$ that preserves the semantic meaning of the source while exhibiting sufficient surface-form variation. Additionally, the system should optionally satisfy user-specified constraints on attributes such as length or lexical diversity.

1.2 Motivation

We chose this problem for several compelling reasons. First, paraphrase generation serves as a foundational capability for numerous downstream applications including data augmentation, information retrieval, question answering, and conversational AI. Second, the task presents a significant technical challenge that requires sophisticated understanding of both semantic and syntactic structures, making it an ideal testbed for advanced NLP techniques. Third, high-quality paraphrase generators can directly improve real-world systems, particularly in query rewriting for Retrieval-Augmented Generation pipelines and multilingual content adaptation. Finally, controllable paraphrase generation with explicit semantic preservation guarantees remains less explored, presenting an opportunity for meaningful contribution. Our project closely follows [1], reproducing and improving upon their approach to precise paraphrase control, with a focus on the more feasible axes of length and lexical diversity control.

2 Related Work

2.1 Sequence-to-Sequence Approaches

Early neural approaches to paraphrase generation adapted sequence-to-sequence architectures with attention mechanisms. These models treated paraphrasing as a translation task from source to target within the same language, demonstrating that neural networks could generate reasonable paraphrases though outputs often lacked diversity and tended to produce conservative reformulations.

2.2 Transformer-Based Models

The advent of transformer architectures revolutionized paraphrase generation capabilities. Models like PEGASUS introduced gap-sentence generation as a pre-training objective, achieving strong performance on abstractive

summarization and paraphrasing tasks. T5 [2] demonstrated through its unified text-to-text framework that paraphrase generation could be effectively framed as a conditional generation task with task-specific prefixes. However, these large-scale models face significant challenges in that they require substantial computational resources for fine-tuning and often produce conservative outputs that closely resemble the input, sacrificing diversity for semantic safety.

2.3 Controllable Generation

Recent work has explored controllable text generation through various mechanisms designed to guide model outputs. Control codes have been introduced to enable steering generation toward specific attributes or styles. For paraphrasing, adversarial learning approaches have been employed to encourage syntactic diversity while preserving semantic content through carefully designed training objectives. Other work has addressed the fundamental semantic drift problem by introducing two-stage approaches consisting of diverse candidate generation followed by semantic filtering [1], highlighting the critical importance of explicit semantic validation in paraphrase generation pipelines.

2.4 Evaluation Challenges

A persistent challenge in paraphrase generation research is evaluation methodology. Automatic metrics like BLEU and METEOR measure surface-form similarity but inadequately capture semantic equivalence, often failing to distinguish between true paraphrases and superficial lexical variations. Recent work has leveraged semantic similarity models based on contextual sentence embeddings [3] to better assess meaning preservation through learned representations. BERTScore [4] provides token-level semantic similarity using BERT embeddings for more fine-grained assessment. However, the field still lacks consensus on comprehensive evaluation protocols that effectively balance semantic fidelity, diversity, and fluency in a unified framework.

3 Dataset

Our experimental setup utilizes carefully selected benchmark datasets to train and evaluate the paraphrase generation system. Compared to our initial plan, we have revised dataset choices to improve feasibility and reduce domain shift between training and evaluation. Using datasets from different domains also helps us check whether the model generalizes beyond a single writing style. This reduces the chance that the system simply learns patterns specific to one dataset.

3.1 Training Data

The primary training dataset is the Quora Question Pairs corpus [6], which contains over 400,000 question pairs from the Quora platform, labeled for semantic equivalence. We utilize the positive pairs, those marked as semantically equivalent questions, as paraphrase examples. Although QQP has a relatively uniform language style that may not support formality control well, it is well-suited for studying length and lexical diversity control. We also hold out a portion of QQP as an additional in-domain test set to reduce domain shift between training and evaluation.

Additionally, we include PAWS (Paraphrase Adversaries from Word Scrambling) [8], which contains 108,463 well-formed paraphrase and non-paraphrase pairs with high lexical overlap. PAWS is particularly valuable for training models to distinguish genuine semantic equivalence from superficial word matching, addressing a common failure mode in paraphrase systems.

3.2 Evaluation Data

For evaluation, we employ the Microsoft Research Paraphrase Corpus (MRPC), which contains 5,801 high-quality sentence pairs from news sources; TwitterPPDB, a human-labeled corpus with 51,524 pairs from Twitter for out-of-domain social media evaluation; and a held-out QQP split reserved for in-domain evaluation, to complement the out-of-domain test sets.

3.3 Preprocessing

Our preprocessing pipeline applies lowercasing and punctuation normalization, removes duplicates and near-duplicates to prevent test leakage, filters pairs with extreme length ratios exceeding 3:1, and splits data into training (80%), validation (10%), and test (10%) sets stratified by source length.

4 Evaluation Plan

Comprehensive evaluation of paraphrase generation requires assessing multiple dimensions simultaneously: semantic fidelity to ensure meaning preservation, lexical and syntactic diversity to ensure useful variation, and fluency to ensure natural language quality.

4.1 Automatic Metrics

Semantic Preservation Metrics. To assess meaning preservation, we employ several complementary approaches:

Cosine Similarity with Sentence-BERT [3]: We compute the cosine similarity between dense vector representations of the source sentence and generated paraphrase using Sentence-BERT embeddings. Specifically, we encode both sentences using the `all-MiniLM-L6-v2` model, which produces 384-dimensional dense vectors that capture semantic meaning through fine-tuned contextualized representations. The cosine similarity is calculated as:

$$\text{CosSim}(s, t) = \frac{\mathbf{v}_s \cdot \mathbf{v}_t}{||\mathbf{v}_s|| \cdot ||\mathbf{v}_t||}$$

where $\mathbf{v}_s$ and $\mathbf{v}_t$ are the sentence embeddings for source and target respectively. This metric captures semantic similarity at the sentence level through contextualized representations that encode both lexical semantics and compositional meaning. Unlike word-overlap metrics, cosine similarity on sentence embeddings can detect

semantic equivalence even when surface forms differ significantly (e.g., "The cat sat on the mat" vs. "A feline rested on the rug"). We establish interpretative thresholds: scores above 0.85 indicate strong semantic preservation (acceptable paraphrases), scores between 0.75–0.85 suggest moderate preservation requiring human verification, and scores below 0.75 indicate insufficient semantic fidelity (likely semantic drift or meaning change). These thresholds are calibrated based on preliminary analysis of the validation set and prior work in semantic textual similarity tasks.

BERTScore [4]: We utilize BERTScore to compute token-level semantic similarity using BERT embeddings. Unlike sentence-level cosine similarity, BERTScore provides fine-grained assessment by matching tokens based on their contextual embeddings, computing precision, recall, and F1 scores. This captures both semantic overlap and structural alignment between source and paraphrase, offering complementary insights to sentence-level metrics.

LLM-as-a-Judge [5]: To complement embedding-based metrics, we employ a large language model (GPT-4) as an automated evaluator for semantic equivalence. Recent work has demonstrated that LLMs can serve as effective evaluators that correlate strongly with human judgments while providing scalable assessment with detailed explanations. We prompt the model with a structured evaluation protocol:

```
"Evaluate if the following paraphrase
preserves the meaning of the source:

Source: [source sentence]
Paraphrase: [generated paraphrase]

Rate semantic preservation on a scale
of 1-5 where:
1 = Completely different meaning
2 = Partially preserves meaning with
    significant information loss
3 = Mostly preserves meaning with
    minor discrepancies
4 = Preserves meaning with negligible
    nuance differences
5 = Perfect semantic equivalence

Provide your response in the format:
Rating: [1-5]
Justification: [brief explanation]"
```

The LLM evaluator provides nuanced judgment that captures subtle semantic distinctions, pragmatic implications, contextual appropriateness, and world knowledge that embedding-based metrics may miss. For instance, it can identify when a paraphrase changes factual claims, introduces unintended implications, or shifts sentiment in ways that pure similarity metrics cannot detect. We aggregate ratings across 500 randomly sampled test instances and compute correlation with both automatic metrics (Sentence-BERT cosine similarity, BERTScore) and human annotations to assess reliability and validity. We also analyze the LLM's justifications qualitatively to understand failure modes and cases where it diverges from other metrics. This multi-faceted approach provides both quantitative scores and qualitative insights into semantic preservation quality.

Surface-Form Diversity Metrics. To measure variation from the source, we calculate inverse BLEU scores (1-BLEU), where lower BLEU indicates greater surface variation while maintaining semantics. We compute Self-BLEU to measure diversity within the set of generated paraphrases for the same source, where lower self-BLEU indicates greater intra-set diversity and reduced redundancy. Additionally, we calculate the Lexical Diversity Ratio as the proportion of non-overlapping tokens between source and paraphrase: $\text{LDR} = 1 - \frac{|\text{tokens}_s \cap \text{tokens}_t|}{|\text{tokens}_s \cup \text{tokens}_t|}$, which provides a normalized measure of lexical variation.

Fluency Metrics. We measure Perplexity using a pre-trained language model (GPT-2) to assess naturalness of generated text, where lower perplexity indicates more fluent and natural-sounding outputs that conform to expected language patterns. We also employ METEOR, which captures word-order preservation and paraphrase detection through stemming and synonymy matching, providing insight into both fluency and semantic adequacy.

Composite Score. To provide unified quality assessment, we compute ParaScore that balances semantic similarity and diversity: $\text{ParaScore} = 0.7 \cdot \text{SemSim} + 0.3 \cdot (1 - \text{BLEU})$, where the 0.7 weighting for semantic preservation reflects its critical importance over diversity in maintaining paraphrase validity.

4.2 Human Evaluation

Recognizing the limitations of automatic metrics, we conduct human evaluation on 200 randomly sampled paraphrases. Two independent annotators rate each output on a 5-point Likert scale across three dimensions: semantic equivalence (meaning preservation), fluency (grammatical correctness and naturalness), and diversity (surface-form variation). We report inter-annotator agreement using Krippendorff's alpha to assess annotation reliability.

4.3 Justification

This multi-metric evaluation framework ensures that semantic metrics prevent meaning drift and validate core paraphrase quality, diversity metrics avoid degenerate solutions where outputs trivially copy inputs, fluency metrics ensure practical usability in downstream applications, and human evaluation validates automatic metrics and captures nuanced quality aspects. The inclusion of LLM-as-a-judge provides an intermediate assessment layer that combines scalability of automatic metrics with nuanced reasoning capabilities, bridging the gap between pure similarity metrics and costly human evaluation.

5 Plans for Methods

5.1 Baseline Methods

We establish baseline performance using three approaches. First, a rule-based system using WordNet for synonym replacement establishes a lower bound. Second, a back-translation pipeline [7] translating English to French and back using pre-trained MarianMT models serves as an unsupervised baseline. Third, a **vanilla T5-base fine-tuned on QQP** without any controllable generation mechanism serves as a stronger neural baseline, allowing us to isolate the contribution of our controllability tokens and semantic reranking components.

5.2 Proposed Approach

Our primary approach combines fine-tuned T5-base [2] with controllable generation tokens and a semantic reranking step. We fine-tune T5-base on our paraphrase datasets, leveraging T5's text-to-text framework through task prefixes such as "paraphrase: [source sentence]".

Compared to our original proposal, we have made two key architectural changes: (1) we remove the multi-level character CNN embedding, as T5 already captures substantial morphology through SentencePiece tokenization and the additional engineering overhead is unlikely to yield clear benefits within the project timeline; and (2) we add an explicit semantic reranking step to better ensure semantic fidelity, directly matching the emphasis of our title.

5.2.1 Controllable Generation with Attribute Tokens

We introduce attribute control tokens that enable users to specify desired characteristics. The input format becomes: Input = [task] [attribute] s, where control tokens specify desired attributes.

We focus on two controllable axes that are feasible within the project scope and well-supported by QQP. For **length**, users can specify `[SHORT]`, `[SAME]`, or `[LONG]` to guide output length relative to the source. For **diversity**, users can specify `[CONSERVATIVE]` or `[CREATIVE]` to balance between safe and exploratory paraphrasing, operationalized via lexical overlap thresholds.

We deliberately exclude *formality control* from our scope. Automatically assigning formality labels can introduce significant noise, and no off-the-shelf, cleanly-labeled formality corpus aligned with our training data is readily available. Length and lexical diversity labels can instead be reliably inferred from source–paraphrase pair statistics (length ratio and token overlap), avoiding the labeling noise associated with formality annotation.

During training, we automatically infer these attributes from ground-truth paraphrases by analyzing their properties and conditioning generation accordingly. This allows the model to learn the relationship between attribute specifications and corresponding linguistic transformations.

5.2.2 Semantic Reranking

To better match the semantic preservation emphasis of our system, we introduce a **semantic reranking step** at inference time. For each input sentence, we generate k diverse paraphrase candidates using diverse beam search [9] and rerank them using a combination of Sentence-BERT cosine similarity and BERTScore F1:

$$\text{Rerank}(t_i) = \lambda \cdot \text{CosSim}(s, t_i) + (1 - \lambda) \cdot \text{BERTScore}(s, t_i)$$

where λ is a tunable hyperparameter (default $\lambda = 0.6$). The candidate with the highest reranking score is returned as the final output. This step provides an explicit semantic validation layer that is lightweight and orthogonal to model training.

5.2.3 Inference Strategy

At inference time, we employ diverse beam search [9] with a diversity penalty to generate k distinct paraphrase candidates. The semantic reranking module then selects the best candidate based on semantic fidelity, while still providing users access to the full candidate set for downstream selection.

6 Timeline

Week	Tasks
9 (3/10-3/16)	Dataset acquisition and preprocessing (QQP + PAWS). Implement baseline methods (rule-based, back-translation). Setup evaluation pipeline with automatic metrics.
10 (3/17-3/23)	Evaluate baselines on all datasets. Setup T5 fine-tuning infrastructure. Train vanilla T5 as strong baseline. Analyze baseline results.
11 (3/24-3/30)	Implement controllable generation with length and diversity attribute tokens. Fine-tune T5 on combined QQP+PAWS datasets. Conduct validation experiments. **Status report due (3/27)**.
12 (3/31-4/6)	Implement diverse beam search for inference. Implement and tune semantic reranking module. Hyperparameter tuning and optimization. Comprehensive evaluation on test sets. Compare all methods with baselines.
13 (4/7-4/13)	Conduct human evaluation study (200 samples). Final analysis and result documentation. Prepare presentation materials. **Project presentation and final report due**.

References

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